

Amrita Vishwa Vidyapeetham
Amrita School of Engineering, Amritapuri
Department of Chemistry
For Research Scholars
SS855 Research Methodology in Applied Sciences
Ph.D Examination, July- 2023

Duration: 3 Hours

Max. Marks: 100

Part -A

(Each question carries 2 marks)

1. Mention tools of qualitative data collection.
2. Define: (a) Systematic sampling (b) TRIPS Agreement
3. Outline the steps involved in initiating research.
4. Mention the requirements of a good table.
5. Discuss the importance of performing a literature survey in research.

Part-B

(Each question carries 7 marks)

6. Detail about the problems faced by women researchers in our country.
7. Detail about how to write a research paper. Give the general structure of a research document.
8. What are the components of the research proposal? How to write a good research proposal?
9. 'A research problem is not solved by apparatus; it is solved in a man's head' -Justify.
10. Differentiate (a) Hypothesis and Theory (b) Conceptual and Empirical Research
(c) Convenience and Judgment sampling
11. "Imagination is more important than knowledge, Knowledge is limited, Imagination encircles the world"- Enumerate.
12. Elaborate on processes involved in selecting and formulating the research problem.
13. Describe in detail the methods involved in filing a patent.
14. Enumerate the role of computers in research. Add on its merits and demerits.
15. Explain: (a) Mathematical Modelling, (b) Trademarks

Part-C

(Each question carries 10 marks)

16. L'Oreal manufactured a new brand of herbal cosmetics. Design a suitable questionnaire to measure customer satisfaction with the product.

17. Find the Rank Correlation: In an aptitude test, two judges rank the 10 competitors in the following order:

<u>Individual</u>	<u>Rank given by Judge I</u>	<u>Rank given by Judge II</u>
1	6	4
2	4	1
3	3	6
4	2	7
5	1	5
6	7	8
7	9	10
8	8	9
9	10	3
10	5	2